

3.a) What is digital divide? How to bridge the digital divide?

Def: Digital divide refers to the gap between individuals who have access to modern information and communication technologies such as computers and internet and those who do not.

There are several types of digital divide as follows:-

(i) Access Divide:

→ This refers to the unequal availability of digital infrastructure and devices due to economic constraints or geographical isolation.

(ii) Use Divide:

→ This highlights the gap between those who possess the skills to navigate digital technologies and those who do not.

(iii) Gender Gap:

→ Women and girls often face barrier for accessing technologies due to cultural norms, safety concerns and economic factors.

(iv) Social Divide:

→ Socio-economic status significantly influences digital inclusion.

Digital divide can be bridged by following ways:-

(i) Improve Infrastructure;

→ By building internet networks and providing electricity in villages and remote areas.

(ii) Make devices affordable;

→ By offering low-cost smartphones, computers and internet plans so more people can afford them.

(iii) Teach digital skills;

→ Starting programs to train people on how to use the internet, apps, and online services.

(iv) Set up community centers;

→ By opening centers where people without internet at home can go online and learn.

(v) Use mobile technology;

→ By promoting the use of smartphones as they are cheaper and more common than computers.

(vi) Supportive Government Policies;

→ Making laws and rules that help everyone to get equal access to technology.

(vii) Create Inclusive Content;

→ By making apps, websites and services that work for everyone, including people with disabilities and or those who speak different languages.

(viii) Public-Private Partnerships;

→ By collaboration between governments, private sector companies and NGOs, can pool resources and expertise to address the digital divide more effectively.

3.6) What are the ^{different} responsibilities of a profession? Discuss.

Professional responsibilities are those additional responsibilities that computing professionals should undertake because of their special knowledge and skill.

The different responsibilities of a profession are as follows:-

(i) Responsibility to Public Welfare;

→ IT professionals must ensure that their work does not harm society.

→ Example: A software engineer should design systems that are safe, secure and protect user privacy.

(ii) Responsibility to maintain Integrity and Honesty;

→ Professionals must be honest about their work, avoid false claims and not engage in plagiarism or data manipulation.

→ Example;

A programmer should not take credit for someone else's code or sell fake software.

(iii) Responsibility to protect Privacy and Confidentiality;

→ Professionals must protect sensitive data, personal information and not misuse or leak it.

→ Example;

DBA must ensure customer data is secure and not shared without permission.

(iv) Responsibility to follow laws and ethical standards;

→ Professionals must follow laws like data protection laws and ethical guidelines of their organization or professional bodies.

→ Example;

Not creating or distributing pirated software or malware.

- (v) Responsibility to Ensure Fairness and Avoid Bias:
- Professionals must ensure that their designs do not discriminate against any group.
 - Example:
Developing AI systems that do not have gender or racial bias.

- (vi) Responsibility of ~~Learn~~ Lifelong Learning:
- Professionals must keep updating their knowledge and skills to serve society better.
 - Example:
Learning about cybersecurity trends to protect users from new threats.

- (vii) Responsibility to contribute to Society:
- Beyond their job, Professionals should work for the betterment of society such as creating accessible systems for disabled users or supporting digital literacy.

4. a) What are the methods of resolving complexity of software?
Can they be resolved completely? Give your understanding.

Ans The methods of resolving complexity of software are as follows:-

(i) Modularity: Break the software into smaller, independent modules or components.

(ii) Abstraction: Hide unnecessary details and show only the essential parts.

(iii) Top-down and Bottom-up design:

- Top-down: Start from the overall system and break it down into smaller parts.

- Bottom-up: Build small components first and then combine them into larger systems.

(iv) Encapsulation: Keep data and functions that work on the data together and hide them from other parts of the system.

(v) Standardization and use of frameworks: Follow coding standards, use well-known frameworks and libraries.

(vi) Good documentation and comments: Writing clear documentation and comments helps make the software easier to understand for others (or yourself later).

(vii) Code reuse: Reuse existing, tested code instead of writing new code for every feature.

No, software complexity cannot be completely removed.

→ This is because some complexity comes from the problem itself, called essential complexity. For example, software for banking or space missions is naturally complex because the tasks are complex.

→ What we can do is manage and reduce accidental complexity i.e. complexity added by poor design, bad coding practices or unnecessary features.

While we can never fully remove complexity from software, we can handle it smartly through good design, planning and coding practices.

4.b) What is privacy protection? What are some of the roles that can be taken by government and computer professionals to protect privacy?

Def: Privacy protection means keeping people's personal data safe from being misused, leaked or accessed without permission.

- It ensures that sensitive information like name, address, financial details or health records are kept secure and used only for the right purpose.
- Privacy protection is important to build trust, respect rights and prevent harm or fraud.

Both government and computer professionals have important roles in protecting people's privacy in the digital world.

(*) Roles of Government:-

(i) Make strong privacy laws;

- The government can create and enforce data protection laws that control how personal data is collected stored and shared.

(ii) Monitor and regulate companies;

- The government can set up regulatory bodies to check whether companies follow privacy rules and punish those who misuse data.

(iii) Promote awareness;

- The government can run campaigns to educate people about their privacy rights and safe online behavior.

(iv) Encourage secure technologies;

- The government can support the development of technologies that protect privacy, like encryption.

(*) Roles of Computer Professionals;

- (i) Design privacy-friendly systems;
 - IT professionals should build systems that collect only necessary data and protect it through security measures like encryption and access control.
- (ii) Follow ethical standards;
 - Computer professionals must follow code of ethics (e.g. ACM/IEEE) that promote privacy, honesty and integrity.
- (iii) Secure data storage and processing;
 - Professionals should ensure data is stored securely and protected from unauthorized access or hacking.
- (iv) Stay updated and aware;
 - IT professionals should keep learning about new privacy risks and solutions so they can protect users effectively.

In short, government protect privacy by making and enforcing laws, regulating companies and spreading awareness. Similarly, computer professionals protect privacy by designing secure, ethical systems that respect user data and consent.

6.a) "Admitting mistakes by the management prevents whistle blowing." How? What other measures can prevent whistle blowing? Discuss.

Sol: Whistle blowing means reporting unethical, illegal or harmful activities happening inside an organization.

A whistleblower is a person who informs authorities, media or the public about wrongdoing in the company like fraud, data misuse, safety violations or corruption.

Admitting mistakes by the management prevents whistle blowing because of the following reasons:

- (i) Problem can be fixed inside:
 - When management admits mistakes, employees believe the problem will be solved within the company, so they don't feel the need to tell outsiders.
- (ii) Creates trust:
 - People trust leaders who are honest about their mistakes. This trust makes employees less likely to report issues outside.
- (iii) Shows good example:
 - When leaders admit mistakes, they show they are responsible. Employees will follow this good example and feel safer sharing problems.
- (iv) Reduces fear:
 - When management is open about mistakes, people aren't scared to speak up. They feel the company will listen, so they don't need to go public.
- (v) Helps the company improve:
 - Admitting mistakes shows the company wants to get better. Employees feel part of the solution and don't feel the need to blow the whistle.

Some other measures can prevent whistle blowing by are as follows:-

- (i) By creating a strong internal reporting system, which allow employees to safely report issues inside the company.
- (ii) By promoting an ethical culture, which encourage honesty, fairness and accountability at all levels of the organization.
- (iii) By ~~repor~~ protecting employees who report concerns, which ensure that employees who speak up are not harassed or treated unfairly.
- (iv) Regular audits and
- (iv) By regular auditing and reviewing, which checks processes regularly, the company can catch and fix problems early.
- (v) By training and awareness programs, which teach staff about ethics, laws and the importance of addressing wrongdoing the right way.

6.b) What do you mean by intellectual property? How has law of Nepal protected its different types?

sgf: Intellectual property is a general term that refers to creations of the mind i.e. ideas, inventions, designs, software, writings etc.

→ It gives the creator legal rights to protect and control how their creation is used.

→ Example:

A company's logo, a new medicine design, a song or a book are all other types of intellectual property.

Nepal has made laws to protect various forms of intellectual property so that creators and businesses get fair rights. The law of Nepal has protected different types of intellectual property by following ways:-

(i) Patent:

→ It is protected by: Patent, Design and Trademark Act, 2022 B.S.

→ Nepal gives a patent right to inventors for their new inventions.

→ The inventor can stop others from making, using, selling their invention without permission.

(ii) Design:

→ It is also protected by: Patent, Design and Trademark Act, 2022 B.S.

→ Designs like new shapes, patterns or decorations of products are protected.

→ The designer can prevent copying of their design.

(iii) Trademark:

- It is also protected by: Patent, Design and Trademark Act, 2022 B.S.
- Names, logos, and symbols that identify products or services are protected.
- Others can't use a registered trademark without permission.

(iv) Copyright:

- It is protected by: Copyright Act, 2059 B.S.
- Books, songs, movies, paintings, computer programs are protected.
- The creator controls how their work is copied, shared or changed.

b) Job and Occupation:

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S.N.	Job	Occupation
(i)	A job is a specific work or duty done for an employee for payment.	An occupation is a broad type of work or profession that a person is trained for.
(ii)	Job refers to the position you hold in a company or organization.	Occupation refers to the kind of work you do, no matter where you work.
(iii)	A job can be changed frequently (you may switch jobs).	An occupation remains the same unless you change your field.
(iv)	A job gives you regular income (salary/wages).	An occupation defines your career or trade, not necessarily tied to one income source.
(v)	Example: Software Developer at XYZ Company.	Example: Software Engineering as an occupation.

Q) Digital forgery:

Digital forgery means creating, altering or copying digital data (such as documents, images, videos, signatures or codes) with the intention to deceive or cheat others.

→ Key points about digital forgery:-

- (i) It involves using computers, software or technology to create fake digital content.
- (ii) Common examples: faking digital signatures, editing official documents, manipulating photos or videos, creating fake IDs.
- (iii) Digital forgery can be used for fraud, identity theft, spreading false information or financial crimes.
- (iv) Digital forgery is a serious cybercrime and is punishable under laws related to cyber security (i.e. Nepal's Electronic Transaction Act, 2063 B.S.).
- (v) IT professionals must work to detect, prevent and report digital forgery to maintain trust in digital systems.